

Practical 10 - Assessing Model Performance

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `SAheart.csv`

This and the following practicals make use of a dataset which are a subset of data from the Coronary Risk-Factor Study (CORIS) baseline survey. This was carried out in three rural areas of the Western Cape in South Africa. The participants were white males aged between 15 and 64, and the binary outcome of interest is whether they had myocardial infarction, as recorded in the `chd` variable. For our purposes, we will assume throughout that the `chd` outcome was ascertained during follow-up after the other variables were measured, but note that this is not how the data were actually obtained.

In this practical we will fit a logistic regression model for the CHD outcome and examine the performance of the fitted model using the techniques discussed in the lecture.

Question 1

Load the dataset into R and fit a logistic regression model for the `chd` outcome with covariates `sbp tobacco ldl famhist obesity alcohol age`. What do the model results suggest about the associations between these covariates and the `chd` outcome?

SOLUTIONS

```
SAheart <- read.csv("SAheart.csv", row.names = 1)
chd_mod <- glm(chd~sbp+tobacco+ldl+factor(famhist)+
               obesity+alcohol+age,
               data=SAheart,
               family="binomial")
summary(chd_mod)
```

```
Call:
glm(formula = chd ~ sbp + tobacco + ldl + factor(famhist) + obesity +
     alcohol + age, family = "binomial", data = SAheart)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.1295997	0.9641558	-4.283	1.84e-05	***
sbp	0.0057607	0.0056326	1.023	0.30643	
tobacco	0.0795256	0.0262150	3.034	0.00242	**
ldl	0.1847793	0.0574115	3.219	0.00129	**
factor(famhist)Present	0.9391855	0.2248691	4.177	2.96e-05	***
obesity	-0.0345434	0.0291053	-1.187	0.23529	
alcohol	0.0006065	0.0044550	0.136	0.89171	
age	0.0425412	0.0101749	4.181	2.90e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 596.11 on 461 degrees of freedom
 Residual deviance: 483.17 on 454 degrees of freedom
 AIC: 499.17

Number of Fisher Scoring iterations: 4

Looking first at the p-values, we see we have evidence of associations between tobacco use, LDL cholesterol, family history and age, with the log odds of CHD, after adjusting for the other covariates.

The coefficients are in the directions we would expect for tobacco use, LDL cholesterol, having a family history of the disease and age.

Question 2

Generate the predicted probabilities of CHD from the fitted model, and create a new variable which forms a 0/1 prediction of the outcome based on whether the predicted probability exceeds 0.5 or not. Based on this, estimate the accuracy of the fitted model.

SOLUTIONS

```
yPred <- 1*(chd_mod$fitted.values>0.5)
# cross tabulate predictions with actual outcomes
table(yPred,SAheart$chd)
```

```
yPred   0   1
      0 255  78
      1  47  82
```

```
# calculate proportion where they agree
mean(SAheart$chd==yPred)
```

```
[1] 0.7294372
```

The proportion of individuals where the prediction is correct (i.e. the accuracy) is 0.729.

Question 3

Install the `pROC` package and use code below to generate the sensitivity and specificity at all possible cut-offs in `coords`.

```
library(pROC)
roc_obj <- roc(SAheart$chd, chd_mod$fitted.values)
coords <- coords(roc_obj, "all", ret = c("threshold", "sensitivity", "specificity"))
head(coords)
```

Use `coords` to produce a plot of sensitivity and specificity against cut-off.

Suppose the covariates were to be used to identify men who had a high risk of a future CHD event in order to give further consultation with advice on lifestyle modifications or possible treatments to reduce future CHD risk. In this context, based on the plot what risk cut-off would you recommend?

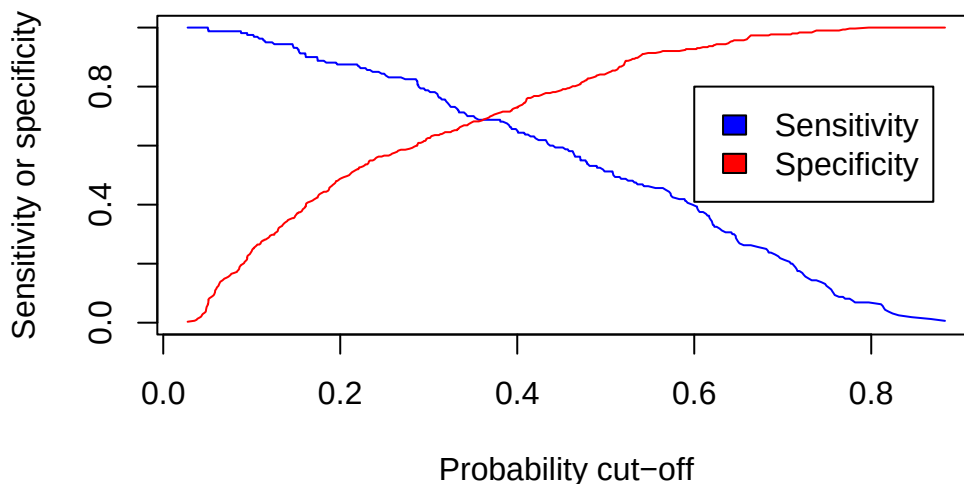
SOLUTIONS

```
library(pROC)
roc_obj <- roc(SAheart$chd, chd_mod$fitted.values)
coords <- coords(roc_obj, "all", ret = c("threshold", "sensitivity", "specificity"))
head(coords)
```

	threshold	sensitivity	specificity
1	-Inf	1	0.000000000
2	0.02753561	1	0.003311258
3	0.03538528	1	0.006622517
4	0.03771336	1	0.009933775
5	0.03854962	1	0.013245033
6	0.04061700	1	0.016556291

We can then plot these with (for example):

```
plot(coords$threshold,coords$sensitivity,
     xlab="Probability cut-off",
     ylab="Sensitivity or specificity",
     type="l", col="blue")
lines(coords$threshold,coords$specificity,
      col="red")
legend(0.6,0.8,legend=c("Sensitivity", "Specificity"),
      fill=c("blue","red"))
```



Given the stated context and intended use, we would probably prefer to keep sensitivity high at the expense of specificity - this way we detect most who will go on to have CHD, with the drawback that we 'flag' for further consultation quite a lot of individuals who were (without further intervention) not going to experience CHD.

Question 4

Examine the `roc_obj` object you have saved and find the estimated AUC. What do you conclude about the predictive ability of the model based on this?

SOLUTIONS

```
roc_obj
```

Call:

```
roc.default(response = SAheart$chd, predictor = chd_mod$fitted.values)
```

Data: chd_mod\$fitted.values in 302 controls (SAheart\$chd 0) < 160 cases (SAheart\$chd 1).
Area under the curve: 0.7816

We obtain an AUC of 0.782. Recalling that 0.5 corresponds to no predictive ability at all and 1 to perfect discrimination, this value would typically be judged as being quite good.

Question 5

Install the `ResourceSelection` package and use the `hoslem.test` function to perform Hosmer-Lemeshow's test with 10 groups. What conclusion do you draw?

SOLUTIONS

```
library(ResourceSelection)
hl_test <- hoslem.test(SAheart$chd, fitted(chd_mod), g = 10)
hl_test
```

Hosmer and Lemeshow goodness of fit (GOF) test

```
data: SAheart$chd, fitted(chd_mod)
X-squared = 8.8323, df = 8, p-value = 0.3566
```

The p-value is again nowhere near significant - we have no evidence of poor fit or poor calibration. We can also examine a table of observed and expected counts to see the values used in the calculation:

```
cbind(hl_test$observed, hl_test$expected)
```

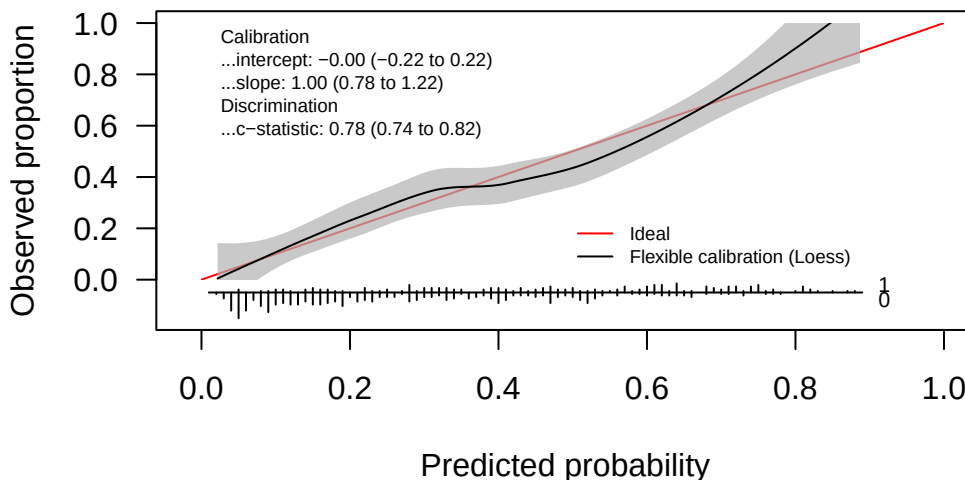
	y0	y1	yhat0	yhat1
[0.0217,0.0683]	45	2	44.55401	2.445989
(0.0683,0.115]	40	6	41.65211	4.347888
(0.115,0.163]	38	8	39.43889	6.561107
(0.163,0.23]	39	7	36.98496	9.015038
(0.23,0.313]	31	15	33.39231	12.607687
(0.313,0.4]	28	18	29.64490	16.355095
(0.4,0.479]	28	18	25.72993	20.270069
(0.479,0.574]	29	17	21.99046	24.009542
(0.574,0.684]	16	30	17.14702	28.852975
(0.684,0.887]	8	39	11.46539	35.534609

Question 6

Install the `CalibrationCurves` package and use the `val.prob.ci.2` function to produce a smooth calibration curve plot. What do you conclude?

SOLUTIONS

```
library(CalibrationCurves)
calPerfObs <- val.prob.ci.2(fitted(chd_mod), SAheart$chd,
  statloc = c(0,0.7),
  cex=0.6,
  cex.leg=0.6)
```



The calibration looks generally good, although the function we have called has not. The 'observed probabilities' (really proportions) are a bit higher than predicted on the right, but we can see there are not so many observations here, so we would expect more variability in the observed proportions.

Question 7

Use the following code to split the sample in training and test subsets.

```
set.seed(592723)
n <- dim(SAheart)[1]
train_idx <- sample(seq_len(n), size = 0.75 * n)
trainData <- SAheart[train_idx, ]
testData <- SAheart[-train_idx, ]
```

Fit the logistic regression using the training subset and estimate the AUC using this subset. Then estimate the AUC of the same fitted model when used to predict in the test subset. For this you will need to use the `predict` function to apply the fitted model to a new dataset.

How do the two AUCs compare, and is it in line with expectations?

SOLUTIONS

```
set.seed(592723)
n <- dim(SAheart)[1]
train_idx <- sample(seq_len(n), size = 0.75 * n)
trainData <- SAheart[train_idx, ]
testData <- SAheart[-train_idx, ]
#to save using the model formula again, we can use update
trainMod <- update(chd_mod, data=trainData)
#AUC in training data is
roc(trainData$chd, trainMod$fitted.values)
```

Call:

```
roc.default(response = trainData$chd, predictor = trainMod$fitted.values)
```

Data: trainMod\$fitted.values in 227 controls (trainData\$chd 0) < 119 cases (trainData\$chd 1)
Area under the curve: 0.7869

```
#AUC when applying fitted model to test data
roc(testData$chd, predict(trainMod,
                          newdata = testData,
                          type="response"))
```

Call:

```
roc.default(response = testData$chd, predictor = predict(trainMod,      newdata = testData, type = "response")
Data: predict(trainMod, newdata = testData, type = "response") in 75 controls (testData$chd == 0)
Area under the curve: 0.7704
```

The AUC estimate from the test data is a bit smaller than that obtained (naively) in the training data, as we would expect. There is though quite a bit of noise/randomness in this because of the random splitting - in the Stata version of this practical the training AUC was 0.7969 while the test was 0.7389. If you want, you can get a sense of this variability by re-running the code with a different seed.

Question 8

Complete the following the following code to perform 10-fold cross-validation in R. What conclusions do you draw?

```
set.seed(26012)
split <- ceiling(runif(n,min=0,max=10))
aucEsts <- array(0, dim=10)
for (i in 1:10) {
  trainData <-
  testData <-
  trainMod <- update(chd_mod, data=trainData)
  aucEsts[i] <- as.numeric(roc(testData$chd, predict(trainMod,
                                                    newdata = testData,
                                                    type="response"))$auc)
}
aucEsts
mean(aucEsts)
```

SOLUTIONS

Our version of the completed code is:

```
set.seed(26012)
split <- ceiling(runif(n,min=0,max=10))
aucEsts <- array(0, dim=10)
for (i in 1:10) {
  trainData <- SAheart[split!=i,]
  testData <- SAheart[split==i,]
  trainMod <- update(chd_mod, data=trainData)
  aucEsts[i] <- as.numeric(roc(testData$chd, predict(trainMod,
```

```

                                newdata = testData,
                                type="response"))$auc)
}
aucEsts

```

```

[1] 0.8500000 0.8076923 0.6907895 0.6521739 0.7544643 0.8885714 0.7352941
[8] 0.7575758 0.6508621 0.8295455

```

```
mean(aucEsts)
```

```
[1] 0.7616969
```

The 10-fold cross validation estimate of AUC is 0.762, slightly lower than the 0.782 we obtained in question 4 when we fitted the model and estimated the AUC using the entire sample. One thing to note is that this cross-validation value is an estimate - if we change the seed, and obtain different partitions of the data, we will obtain different estimates (see below). Moreover, the actual quantity being estimated changes with the number of folds used. See Section 7.10.1 of the Elements of Statistical Learning book if you are interested to read further.

```

#new seed
set.seed(94235)
split <- ceiling(runif(n,min=0,max=10))
aucEsts <- array(0, dim=10)
for (i in 1:10) {
  trainData <- SAheart[split!=i,]
  testData <- SAheart[split==i,]
  trainMod <- update(chd_mod, data=trainData)
  aucEsts[i] <- as.numeric(roc(testData$chd, predict(trainMod,
                                newdata = testData,
                                type="response"))$auc)
}
aucEsts

```

```

[1] 0.7933333 0.7578125 0.8793103 0.6789474 0.9204244 0.7215007 0.8299663
[8] 0.6166329 0.6923077 0.7151515

```

```
mean(aucEsts)
```

```
[1] 0.7605387
```